

Home Lab Exercises for Network Analysis

Lab 1: Encapsulation Analysis

Capture HTTP traffic with Wireshark.

Identify headers at each layer.

Lab 2: ARP Spoofing Detection

Simulate ARP poisoning in a controlled lab.

Observe MAC address changes in packet captures.

Lab 3: Routing and Traceroute

Trace paths and correlate hops with routing decisions.

Identify ICMP message types.

Lab 4: TCP vs UDP Behavior

Compare TCP handshake vs UDP exchange in PCAPs.

Identify retransmissions and dropped packets.

Lab 5: DNS Traffic Analysis

Inspect DNS queries and responses.

Detect anomalous query patterns.

Lab 6: HTTPS Inspection

Analyze TLS handshake metadata.

Differentiate encrypted payload vs visible headers.